

Angles, polygons, congruence and vectors - Set 2 - Solutions

Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19.

Answer key

Question	Answer	Topic
1	D	Angles, polygons, congruence and vectors
2	D	Angles, polygons, congruence and vectors
3	E	Angles, polygons, congruence and vectors
4	C	Angles, polygons, congruence and vectors

Worked solutions

Question 1 GEO-04 · Angles, polygons, congruence and vectors · spec M5.19

In triangle OAB , $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$. M is the midpoint of AB and P is the point on OM with $OP : PM = 2 : 1$. Express $\vec{AP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Answer: D

$$\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b}), \vec{OP} = \frac{2}{3}\vec{OM} = \frac{1}{3}(\mathbf{a} + \mathbf{b}). \vec{AP} = \vec{OP} - \mathbf{a} = \frac{1}{3}\mathbf{b} - \frac{2}{3}\mathbf{a}.$$

Full method - how and why

Vector geometry: build a route from A to P out of the known vectors, using the ratio to fix the fraction.

- 1 Find $\vec{OM}$: M is the midpoint of AB , so $\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$.

why: The midpoint of AB is the average of the two position vectors.

- 2 Use the ratio $OP : PM = 2 : 1$: P is $\frac{2}{3}$ of the way from O to M , so $\vec{OP} = \frac{2}{3} \times \frac{1}{2}(\mathbf{a} + \mathbf{b}) = \frac{1}{3}(\mathbf{a} + \mathbf{b})$.

why: $2 : 1$ means 2 parts out of 3 - convert the ratio into a fraction of the whole.

- 3 Travel from A to P via O : $\vec{AP} = \vec{AO} + \vec{OP} = -\mathbf{a} + \frac{1}{3}(\mathbf{a} + \mathbf{b})$.

why: $\vec{AO} = -\vec{OA} = -\mathbf{a}$; reversing a vector negates it.

- 4 Simplify: $-\mathbf{a} + \frac{1}{3}\mathbf{a} + \frac{1}{3}\mathbf{b} = \frac{1}{3}\mathbf{b} - \frac{2}{3}\mathbf{a}$.

why: Collect the $\mathbf{a}$ terms: $-1 + \frac{1}{3} = -\frac{2}{3}$.

Where it goes wrong

- Using $\frac{1}{2}$ instead of $\frac{2}{3}$ for the ratio point.
- Forgetting the $-\mathbf{a}$ and giving $\vec{OP}$ instead of $\vec{AP}$ - an offered option.

Takeaway Route vectors: $\vec{XY} = \vec{XO} + \vec{OY}$, and a ratio $m : n$ means the fraction $\frac{m}{m+n}$.

Question 2 GEO-06 · Angles, polygons, congruence and vectors · spec M5.14

A ship sails from P on a bearing of 060° for 10 km to Q , then on a bearing of 150° for 10 km to R . What is the bearing of P from R ?

Answer: D

The turn at Q is 90° and $PQ = QR$, so triangle PQR is right-angled isosceles and PR makes 45° with QR . Bearing of R from P is $060 + 45 = 105^\circ$; the back bearing is $105 + 180 = 285^\circ$.

Full method - how and why

A bearings problem with a right angle. Find the shape of the triangle, then work out the bearing and reverse it.

- 1 The bearing changes from 060° to 150° , a turn of 90° , so the path turns through a right angle at Q .

why: The interior angle PQR is therefore 90° .

- 2 Both legs are 10 km, so triangle PQR is right-angled **isosceles**, with the two base angles 45° each.

why: Equal legs give equal base angles, each $\frac{180-90}{2} = 45^\circ$.

- 3 Bearing of R from P : the direction PQ is 060° , and PR is 45° further clockwise, so $060 + 45 = 105^\circ$.

why: R lies on the far side of PQ from north, so add.

- 4 Reverse for the bearing of P from R : add 180° to get 285° .

why: Back bearings differ by 180° (add if under 180° , subtract if over).

Where it goes wrong

- Giving 105° , which is the bearing of R from P - the reverse of what is asked (and an offered option).
- Subtracting 180° and getting a negative, or forgetting three-figure form.

Takeaway Bearings are clockwise from north; a back bearing differs by exactly 180° .

Question 3 GEO-11 · Angles, polygons, congruence and vectors · spec M5.19

$\vec{AB} = 3\mathbf{p} + 2\mathbf{q}$ and $\vec{BC} = k\mathbf{p} - 4\mathbf{q}$, where $\mathbf{p}$ and $\mathbf{q}$ are not parallel. Given that A , B and C are collinear, find k .

Answer: E

Collinear means $\vec{BC} = \lambda\vec{AB}$: $-4 = 2\lambda \Rightarrow \lambda = -2$, so $k = 3\lambda = -6$.

Full method - how and why

Collinear points mean one vector is a scalar multiple of the other. Compare coefficients of the two independent directions.

- 1 Collinear A , B , C means $\vec{BC} = \lambda\vec{AB}$ for some scalar λ .
why: They share the point B , so the two vectors must point along the same line.
- 2 Write it out: $k\mathbf{p} - 4\mathbf{q} = \lambda(3\mathbf{p} + 2\mathbf{q})$.
why: Because $\mathbf{p}$ and $\mathbf{q}$ are not parallel, coefficients can be compared independently.
- 3 Compare $\mathbf{q}$: $-4 = 2\lambda$, so $\lambda = -2$.
why: Use the equation that contains no unknown besides λ .
- 4 Compare $\mathbf{p}$: $k = 3\lambda = -6$.
why: Substitute the value of λ .

Where it goes wrong

- Taking $\lambda = 2$ and answering 6 - an offered option.
- Comparing $\mathbf{p}$ with $\mathbf{q}$ coefficients.

Takeaway Non-parallel vectors are independent: equate their coefficients separately to solve for scalars.

Question 4 GEO-15 · Angles, polygons, congruence and vectors · spec M5.2

A regular hexagon and a regular octagon share a common side. What is the size of the angle between their two other sides that meet at one end of the common side (the angle outside both polygons)?

Answer: C

Angles at the shared vertex: 120° (hexagon) + 135° (octagon) + $x = 360^\circ$, so $x = 105^\circ$.

Full method - how and why

Angles at a point sum to 360° . Find each polygon's interior angle and subtract.

- 1 Hexagon interior angle: exterior $= \frac{360}{6} = 60^\circ$, so interior $= 120^\circ$.
why: Regular polygon: interior $= 180 - \frac{360}{n}$.
- 2 Octagon interior angle: exterior $= \frac{360}{8} = 45^\circ$, so interior $= 135^\circ$.
why: Same formula with $n = 8$.
- 3 At the shared vertex three angles meet and fill a full turn: $120 + 135 + x = 360$.
why: The two interior angles plus the gap between the outer sides make a complete revolution.
- 4 Solve: $x = 360 - 255 = 105^\circ$.
why: That gap is the angle asked for.

Where it goes wrong

- Using 180° instead of 360° at the vertex.
- Using exterior angles in the sum, giving 255° or similar.

Takeaway Regular n -gon interior angle $= 180 - \frac{360}{n}$; angles round a point total 360° .